

vector !lightcolor = (1.Of,1.Of,1.Of)

vector !objectcolor = (0.Of,0.Of,1.Of)

vector !result = lightcolor · objectcolor = (0.Of,0.Of,1.Of)

Where we have written colours written in RGB space, what we see due to the reflection of this light ray from the object will be a scalar product of the two vectors (0.Of, 0.Of, 1.Of) which is Blue! Note that colours also have an alpha coordinate (transparency), that we have omitted here for simplicity. This can be included as the fourth parameter. The simple calculation above shows that the object absorbs a large part of the white spectrum and only a reflects the blue part. Objects can interact in a much more complicated way with light than this and hence it is important to understand this interaction for us to create well-lit scenes.

Lighting Models

Since lighting in the real world is very complex, we approximate them using simplified models that is easier to understand. One such simple model is the Phong Lighting Model (although remember that Blender uses a much more complex model). Another such model is the Gouraud Shading Model. We wouldn't go into details of these lighting models since understanding them would require substantial knowledge of Transformations and Linear Algebra. The Phong Lighting Model (and perhaps the most used one) majorly consist of three components:

1. **Ambient Lighting:** Lighting is never completely dark. Even when all the lights are switched off, and it is dark, there is always some faint light present in the world. This is called ambient light.
2. **Diffuse Lighting:** Simulates the directional impact light has on an object. This creates the most visual part of this model, and it represents most of the reflected light from the faces of the object.
3. **Specular Lighting:** Kind of brings a shininess to the object. Recall, how when you shine a bright light on a metal or some other shiny surface, you get a glare off it. That is modelled by specular light.

Each light source and each object has ambient, diffuse, specular properties which compositely decide what fraction and colour of the ambient, diffuse and specular lights will be reflected and how much absorbed by the object. Note that the lighting effects interact intimately with the texturing, so the desired effect of say how shiny a teapot is, may be obtained by tweaking the textures instead of the lighting itself.